

Mariem JERBI

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[LinkedIn](#) | [Portfolio](#)

Profile

Final-year engineering student in **Data Engineering and Decision Systems**, specialized in **Data Science, NLP, and Generative AI (LLMs, RAG)**. Experienced in designing **end-to-end intelligent systems**, from data analysis to application integration, with measurable performance. Seeking an **end-of-studies internship in Data Science / Artificial Intelligence**.

Education

National School of Electronics and Telecommunications of Sfax (ENET'Com)

Engineering Degree – Data Engineering and Decision Support Systems 2023 – 2026 (*expected*)

Faculty of Sciences of Sfax (FSS)

Preparatory Cycle in Mathematics and Physics (MP) 2021 – 2023

Lycée Tayeb Mhiri – Sfax

Baccalaureate in Mathematical Sciences 2021

Professional Projects

PsyBot – RAG-based Conversational Chatbot

Summer Internship

- Design and development of an intelligent chatbot based on **Retrieval-Augmented Generation (RAG)** to provide contextual and personalized responses.
- Implementation of a **hybrid retrieval approach** combining **BM25** and **vector search (Qdrant)**, with a **reranking** mechanism.
- Generation of semantic embeddings using **BGE-M3** to improve response relevance.
- Integration of **LLMs via Ollama** within a scalable web architecture.
- **Technologies:** React.js, Node.js, Firebase, Ollama, Qdrant, BM25, BGE-M3.

Academic and Personal Projects

Spam Detection System (NLP & Deep Learning)

- Development of an **LSTM model** for spam message classification achieving **98% accuracy**.
- Advanced text preprocessing and hyperparameter optimization.
- **Technologies:** Python, TensorFlow/Keras, NLP, NLTK.

SAVI-AI – Arabic Violent Speech Detection

- Design of a text classification system for Arabic content into *violent, non-violent, or neutral*.
- Training and comparison of **Naïve Bayes** and **AraBERT** models.
- Integration of a reactive chatbot providing real-time feedback to users exhibiting harmful linguistic patterns, promoting positive communication.
- **Technologies:** Python, TensorFlow, AraBERT, LLMs, Flask, Flutter.

Sahha AI – Medical NLP Diagnostic System for Tunisian Dialect

- Design of an NLP system based on **Transformers** to translate Tunisian dialect (Derja / Arabizi) into standard medical French.
- Implementation of a **vector semantic inference** approach to estimate disease probabilities from symptoms.
- Use of **Gemini 2.5 Flash** and comparison with **GPT-4, Claude, and Qwen** in terms of linguistic robustness, latency, and cost.
- System evaluation using quantitative metrics such as **F1-score, Top-1 Accuracy, and BLEU Score**.
- **Technologies:** Transformers, LLMs, NLP, Embeddings, Cosine Similarity.

Technical Skills

Programming Languages: Python, Java, C, JavaScript, PHP, Scala

Data Science & AI: Machine Learning, Deep Learning, NLP, Transformers, LLMs, RAG, Embeddings

Frameworks & Tools: TensorFlow, Flask, Flutter, React.js, Node.js

Databases: MySQL, MongoDB

Environments: Jupyter Notebook, Google Colab, VS Code

Certifications and Training

- Microsoft Azure AI Fundamentals
- Python & Machine Learning Training
- Python / SQL / SQLite Training
- Generative AI and Computer Vision Workshop
- Flutter Development Workshop
- Participation in a Web Development Hackathon

Languages

Arabic (Native)

English (B2)

French (B1)

German (A2)